The current in an AC circuit is said to be wattless current if the power consumed by it is zero. This wattless current happens in a purely inductive or capacitive circuit.

4.8.3 Power factor

The power factor of a circuit is defined in one of the following ways:

(i) **Power factor = cos ϕ = cosine of the angle of lead or lag**

(ii) **Power factor** $= \frac{R}{Z} = \frac{\text{Resistance}}{\text{Impedance}}$

(iii) **Power factor** $= \frac{VI\cos\phi}{VI}$

$= \frac{\text{True power}}{\text{Apparent power}}$

Some examples for power factors:

(i) Power factor = cos 0° = 1 for a pure resistive circuit because the phase angle ϕ between voltage and current is zero.

(ii) Power factor $= \cos\left(\pm\pi/2\right) = 0$ for a purely inductive or capacitive circuit because the phase angle ϕ between voltage and current is $\pm\pi/2$.

(iii) Power factor lies between 0 and 1 for a circuit having R, L and C in varying proportions.

4.8.4 Advantages and disadvantages of AC over DC

There are many advantages and disadvantages of AC system over DC system.

Advantages:

(i) The generation of AC is cheaper than that of DC.

(ii) When AC is supplied at higher voltages, the transmission losses are small compared to DC transmission.

(iii) AC can easily be converted into DC with the help of rectifiers.

Disadvantages:

(i) Alternating voltages cannot be used for certain applications e.g. charging of batteries, electroplating, electric traction etc.

(ii) At high voltages, it is more dangerous to work with AC than DC.

EXAMPLE 4.26

A series RLC circuit which resonates at 400 kHz has 80 μH inductor, 2000 pF capacitor and 50 Ω resistor. Calculate (i) Q-factor of the circuit (ii) the new value of capacitance when the value of inductance is doubled and (iii) the new Q-factor.

Solution

$L = 80 \times 10^{-6}$H; $C = 2000 \times 10^{-12}$ F

$R = 50\ \Omega$; $f_r = 400 \times 10^3$Hz

(i) Q-factor, $Q_1 = \frac{1}{R}\sqrt{\frac{L}{C}}$

$= \frac{1}{50}\sqrt{\frac{80\times10^{-6}}{200\times10^{-12}}} = 4$

(ii) When $L_2 = 2$ L

$= 2 \times 80 \times 10^{-6}$ H $= 160 \times 10^{-6}$ H,

$C_2 = \frac{1}{4\pi^2 f_r^2 L_2}$

$= \frac{1}{4\times3.14^2\times\left(400\times10^3\right)^2\times160\times10^{-6}}$

$\simeq 1000\times10^{-12}\,F$

$C_2 \simeq 1000\,pF$

(iii) $Q_2 = \frac{1}{R}\sqrt{\frac{L_2}{C_2}} = \frac{1}{50}\sqrt{\frac{160\times10^{-6}}{1000\times10^{-12}}}$

$= \frac{1}{50}\sqrt{\frac{16\times10^{-5}}{10^{-9}}} = \frac{4\times10^2}{50} = 8$

UNIT-4(XII-Physics_Vol-1).indd 266 04-03-2019 10:41:33